26900160 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCCFS301 - SECURITY ANALYSIS USING PYTHON

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer any **10** questions

Each question carries **1** marks

1. Explain why lists are called mutable objects in Python. [U] / [CO1]
2. Explain how Python evaluates an expression. [U] / [CO1]
3. Distinguish between in and not in operators when used with strings. [U] / [CO1]
4. Differentiate between for and while loops. [U] / [CO2]
5. Define Constructor. [K] / [CO2]
6. Define recursion. [K] / [CO2]
7. Differentiate between local and global variables. [U] / [CO2]
8. List the basic components of a function definition. [K] / [CO2]
9. Describe the difference between an expression and a statement in Python. [U] / [CO1]

10. Express the role of SMTP in email communication. [U] / [CO3]

11. Describe the steps to Develop an Efficient IP Scanner in Python. [An] / [CO3]

12. Explain what is ping sweep? [An] / [CO3]

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

13. Explain how to create and access elements of a dictionary with suitable examples. [U] / [CO1]

14. Illustrate the procedure for creating and executing a simple Python program. [U] / [CO1]

15. Illustrate the flow of execution of a simple Python program. [U] / [CO2]

16. Explain Relational Operators with an example. [U] / [CO2]

17. Differentiate between a module and a package. [U] / [CO2]

18. Outline the client-server model used in socket programming [U] / [CO3]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

19. Discuss sets in Python and explain their creation, properties, and common operations. [U] / [CO1]

20. List common built-in exceptions and explain how exceptions are handled. Describe the method of defining a new exception in Python. [U] / [CO2]

21. Discuss the scope and objectives of penetration testing with examples [U] / [CO3]

[10x2 = 20]